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Evans, II

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(54) **PAYLOAD DELIVERY AND DROP DEVICE FOR UNMANNED AERIAL VEHICLE FOR AUTOMATIC RELEASE UPON SURFACE CONTACT**

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(51) **Int. Cl.**
B64D 1/12 (2006.01)

B64U 101/64 (2023.01)

(52) **U.S. Cl.**
CPC **B64D 1/12** (2013.01); **B64U 2101/64** (2023.01)

(58) **Field of Classification Search**
CPC B64D 1/12; B64U 2101/64
See application file for complete search history.

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Primary Examiner — Christopher P Ellis

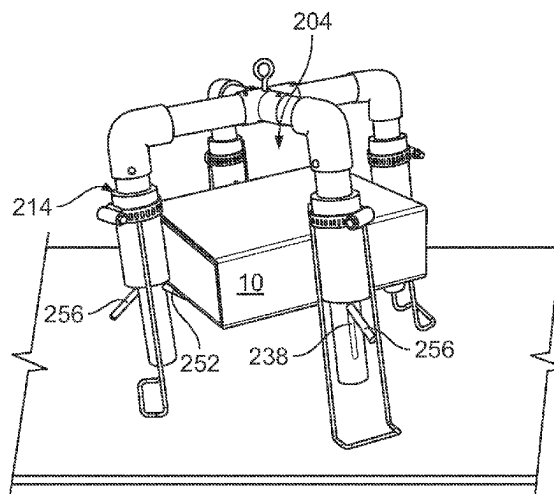
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(57) **ABSTRACT**

Embodiments of a payload delivery and drop device for an unmanned aerial vehicle, and method of its use are described. An apparatus includes a moveable collar defining a lumen having a centerline, and a rigid bar that has a first portion and a second portion. The first portion of the rigid bar has a centerline and the second portion of the rigid bar has a centerline. The second portion of the rigid bar is disposed within the lumen of the collar when the rigid bar is in a first position. The centerline of the first portion of the rigid bar is substantially non-parallel to the centerline of the lumen of the moveable collar when the rigid bar is in the first position. The centerline of the second portion of the rigid bar is substantially non-parallel to the centerline of the lumen when the rigid bar is in the second position.

11 Claims, 11 Drawing Sheets

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Related U.S. Application Data

continuation of application No. 17/200,686, filed on
Mar. 12, 2021, now Pat. No. 11,225,325.

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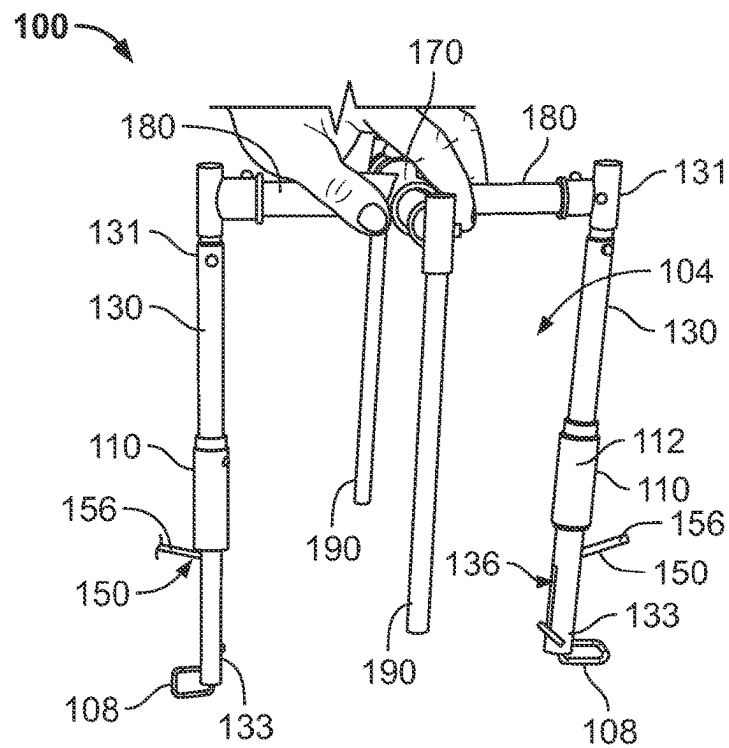


FIG. 1

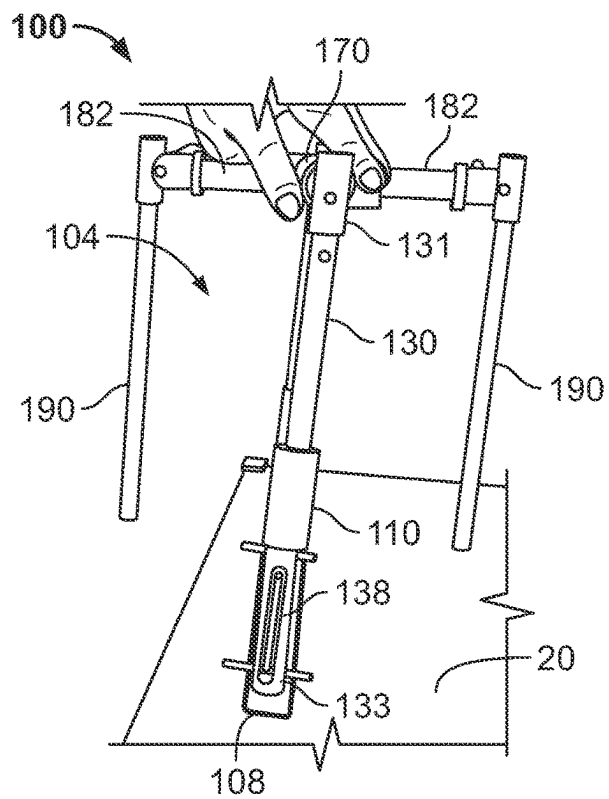


FIG. 2

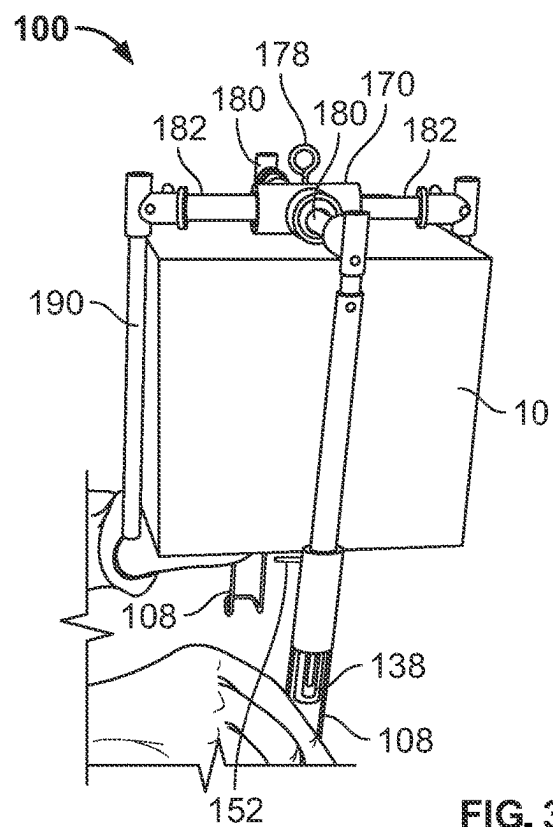


FIG. 3

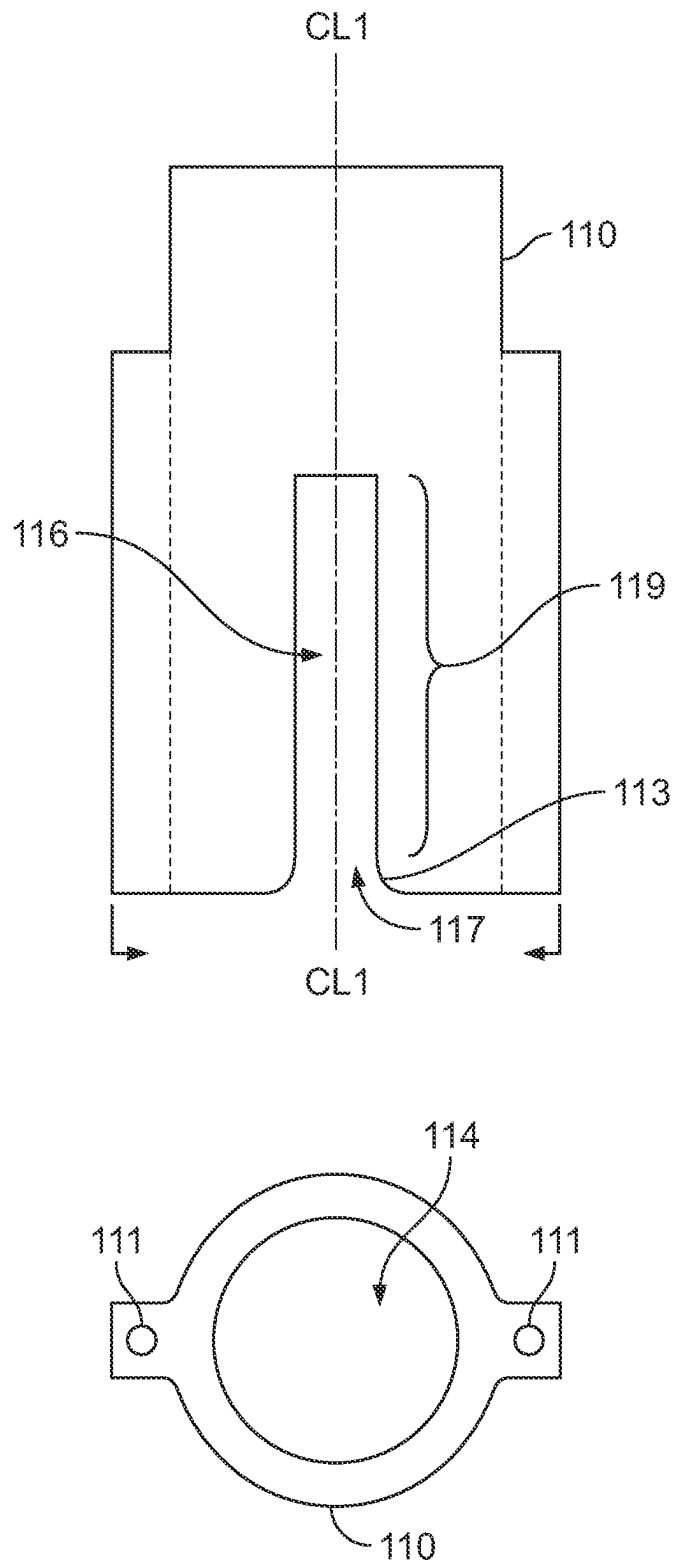


FIG. 4

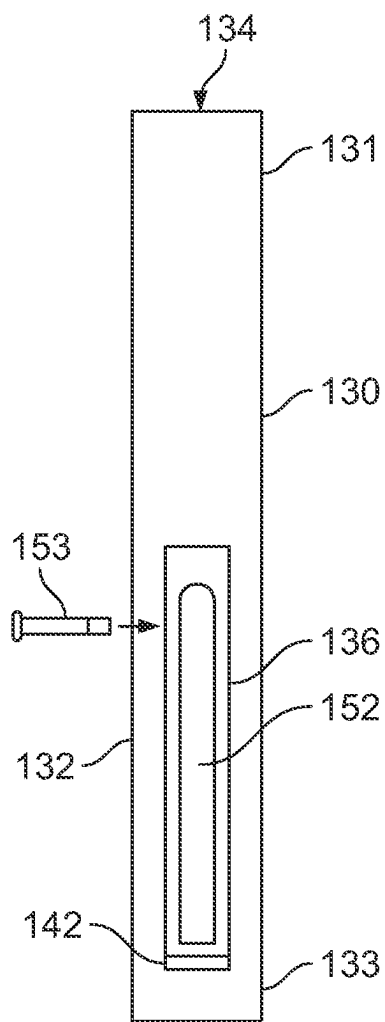


FIG. 5

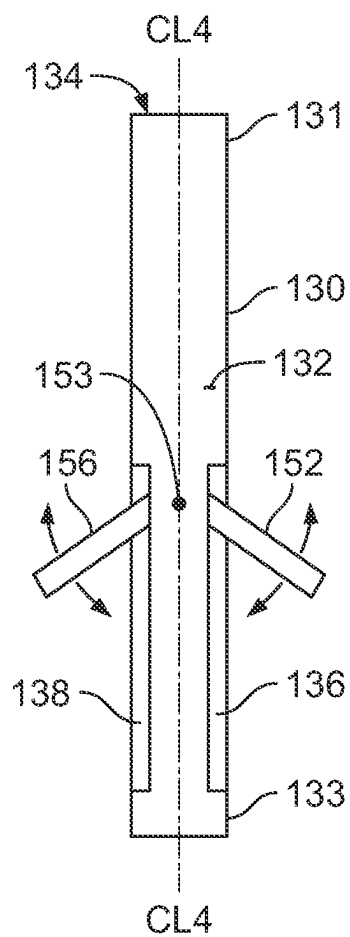


FIG. 6

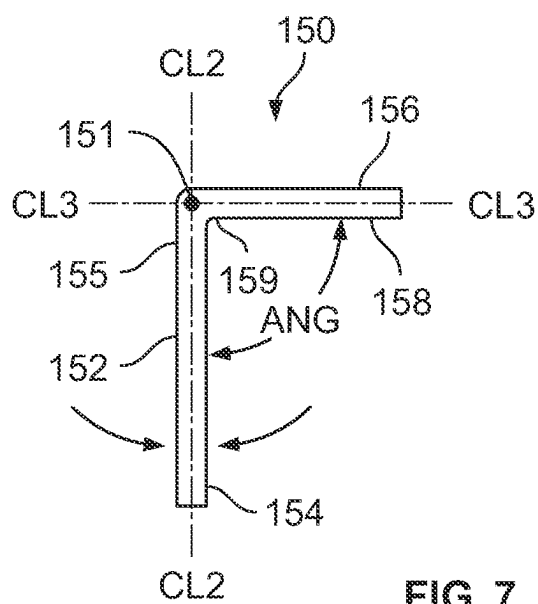


FIG. 7

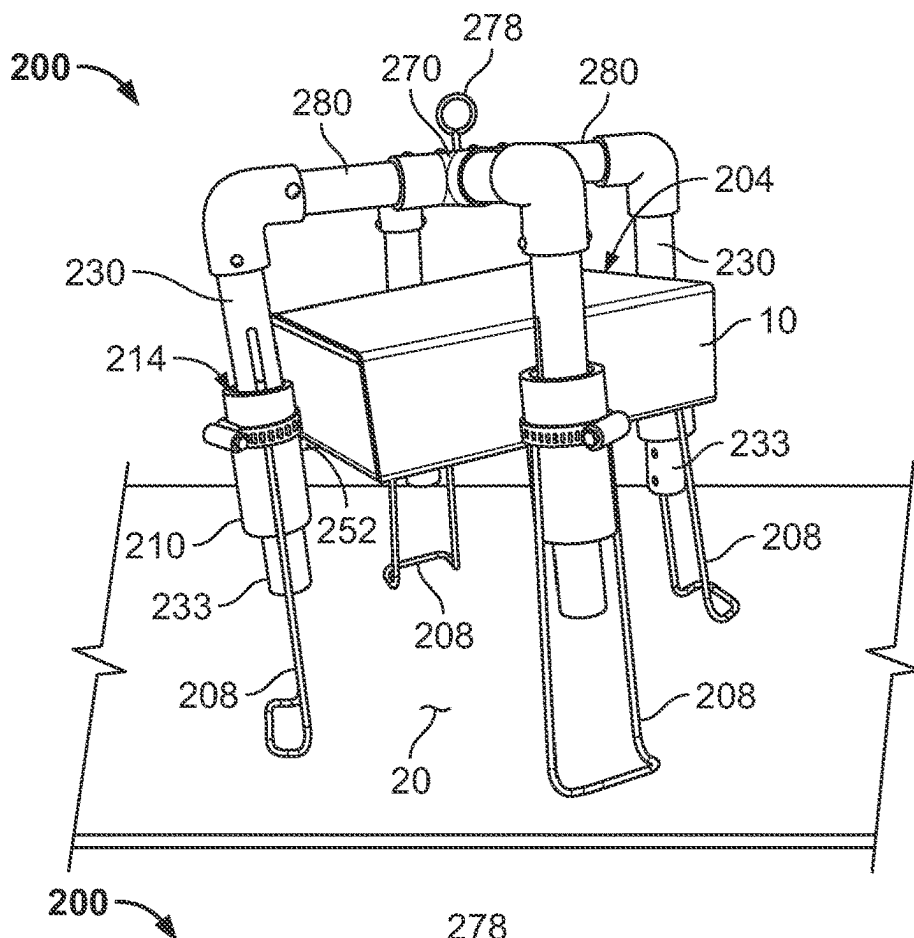


FIG. 8

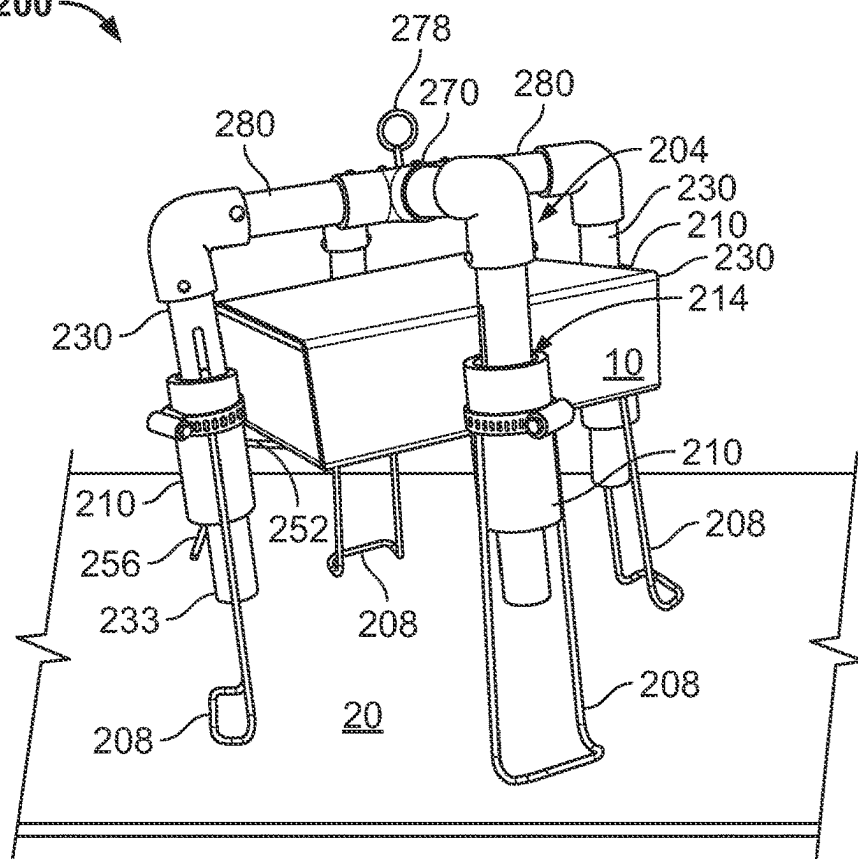


FIG. 9

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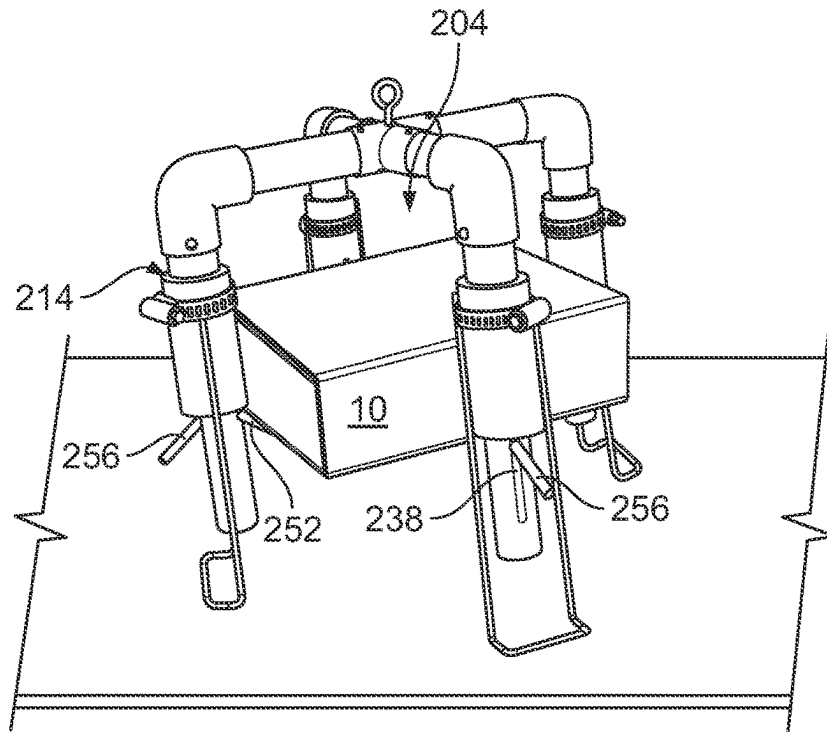


FIG. 10

200

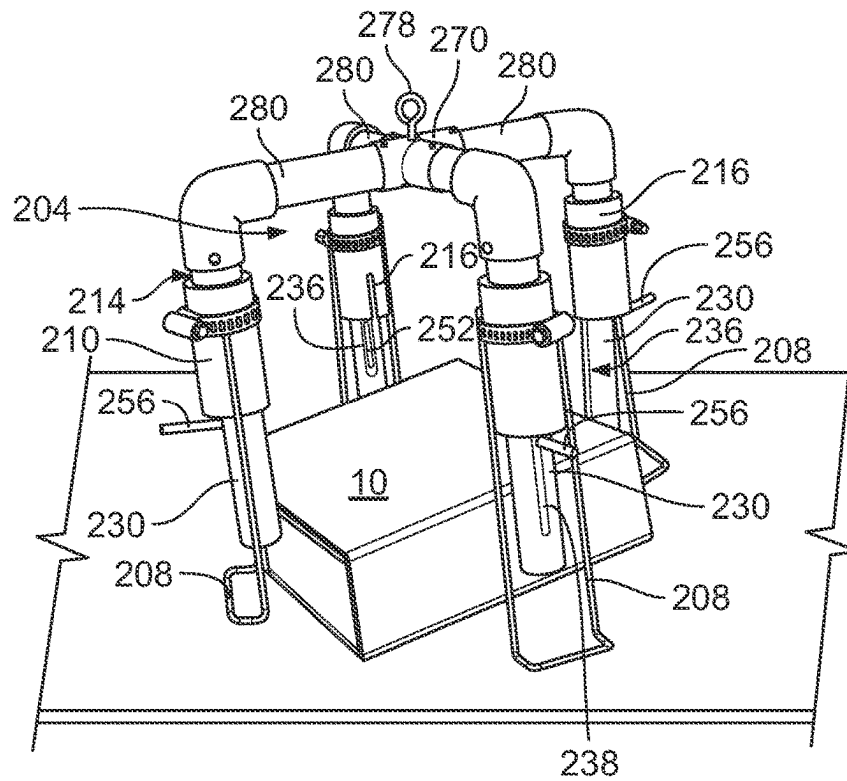


FIG. 11

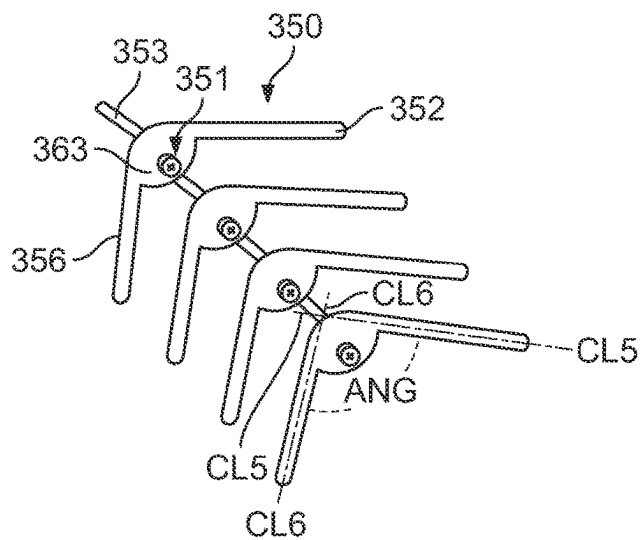


FIG. 12

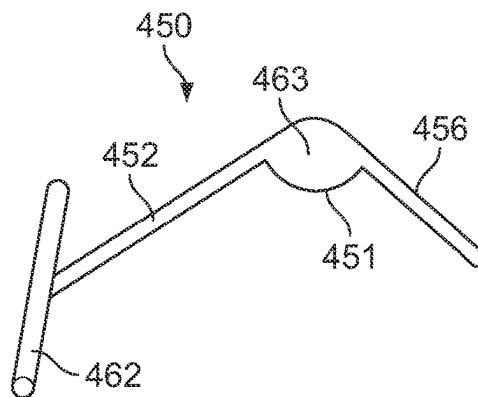


FIG. 13

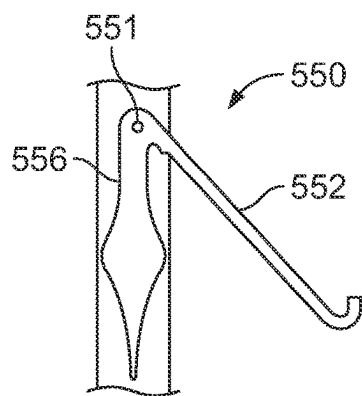


FIG. 14

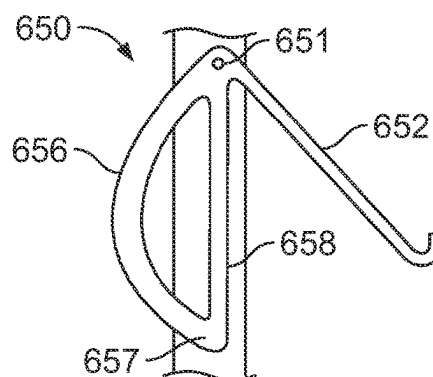


FIG. 15

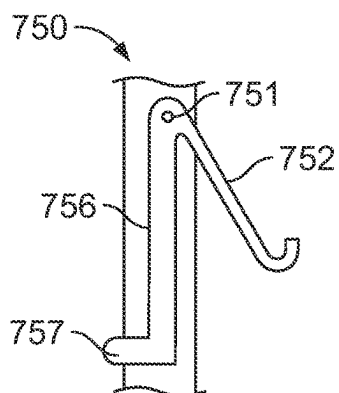


FIG. 16

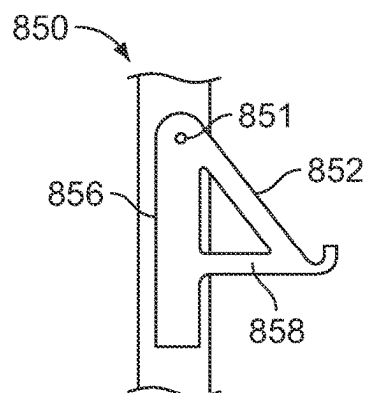


FIG. 17

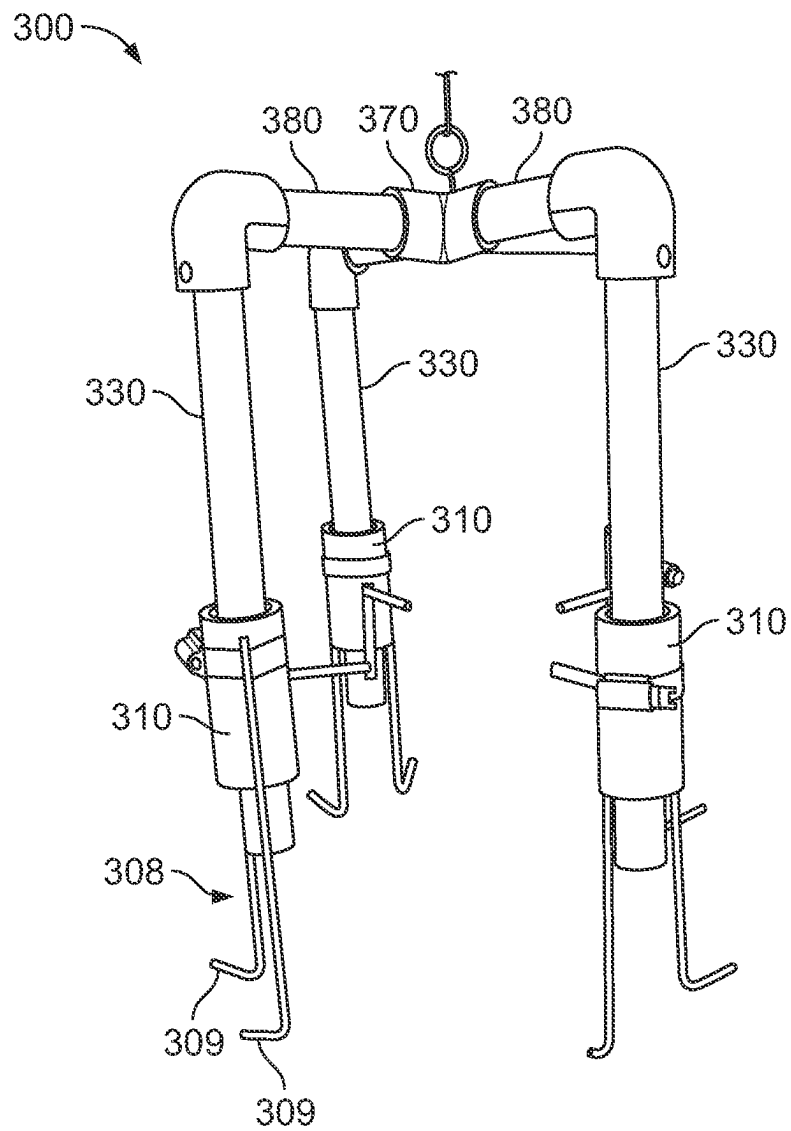


FIG. 18

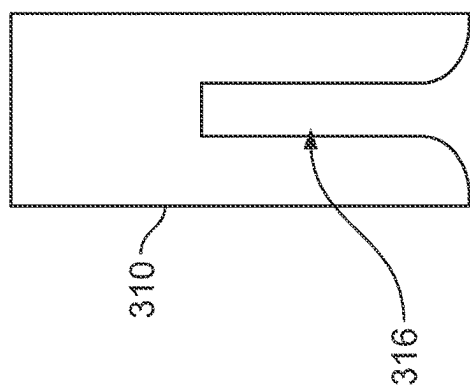


FIG. 19

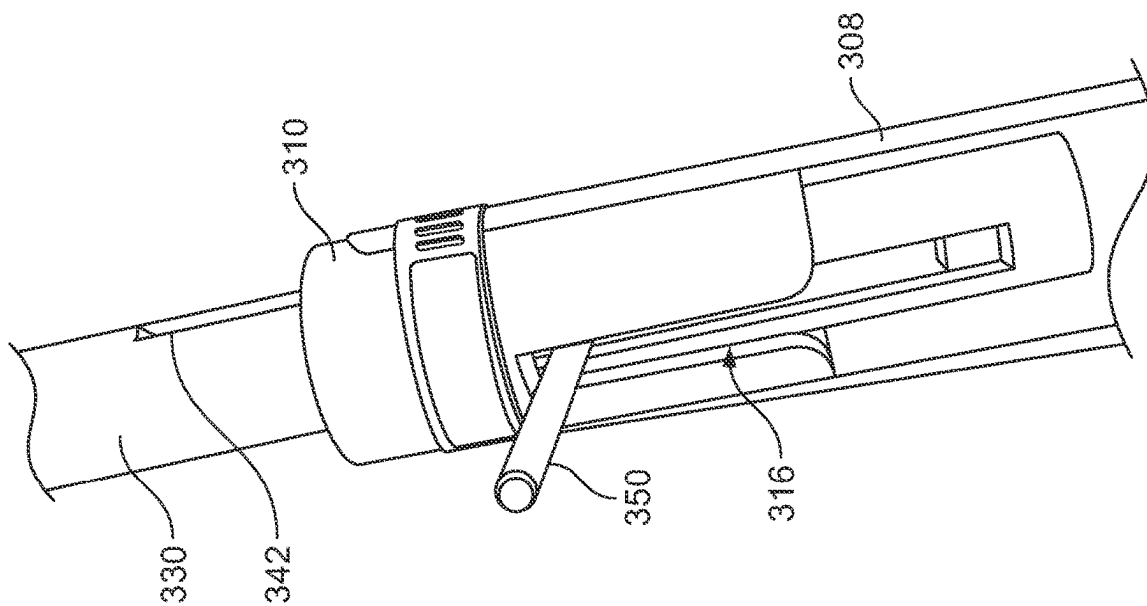


FIG. 20

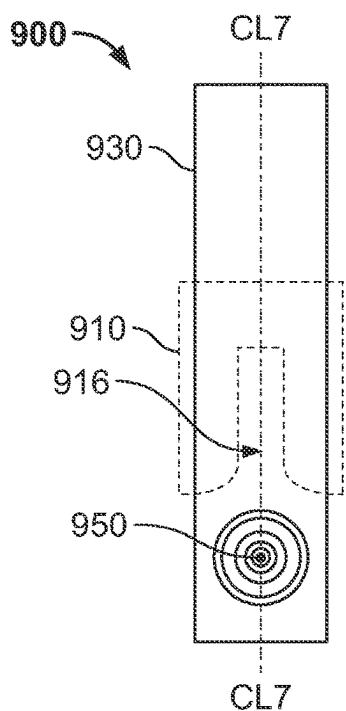


FIG. 21

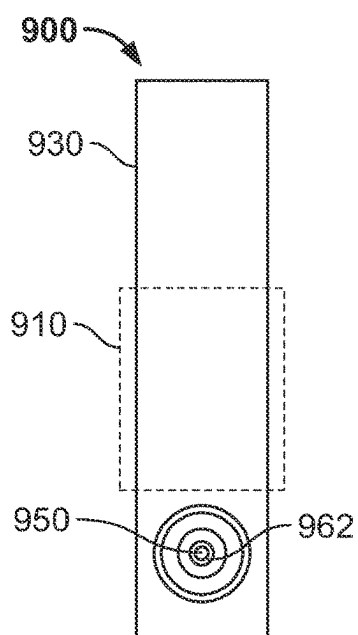


FIG. 22

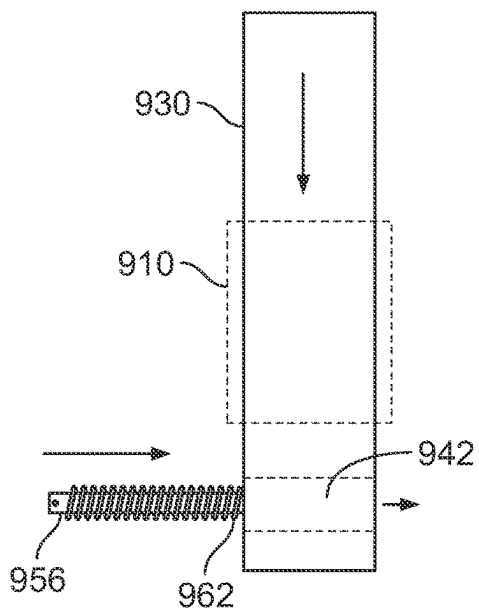


FIG. 23

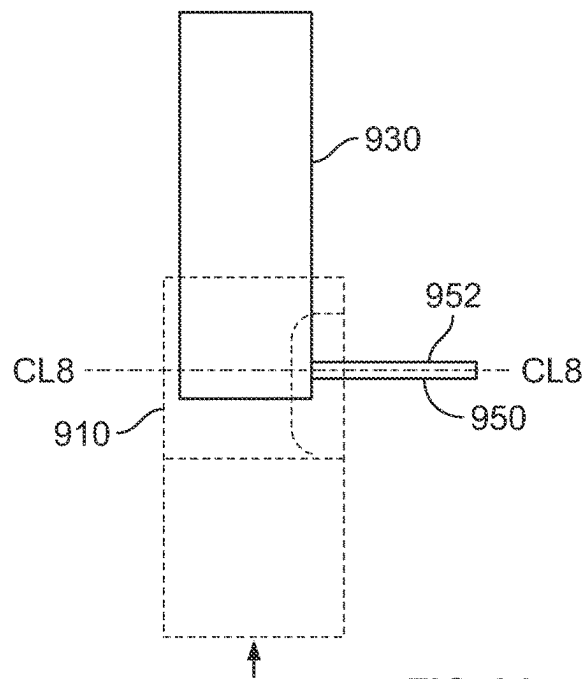


FIG. 24

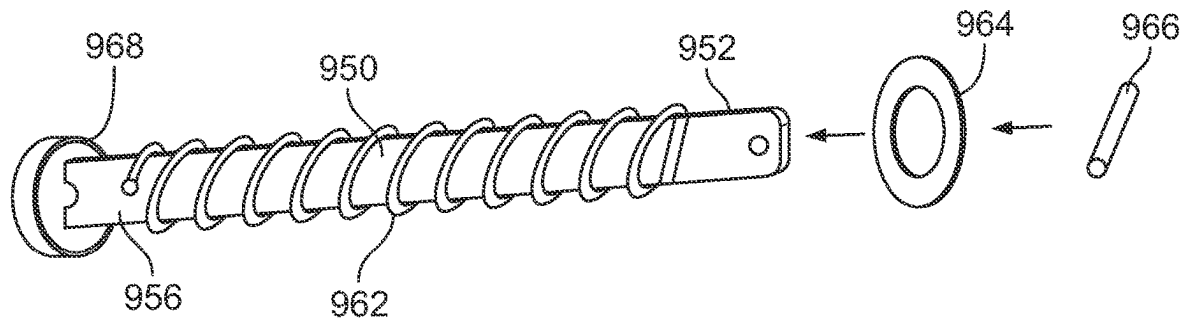


FIG. 25

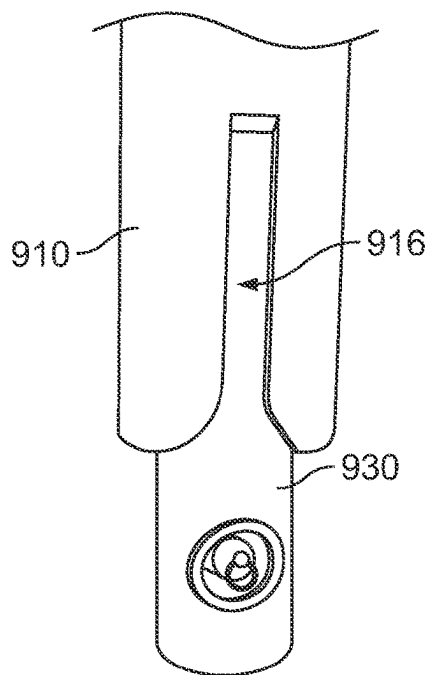


FIG. 26

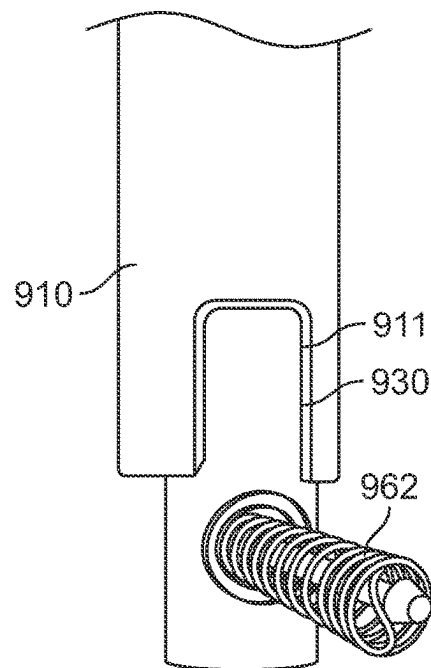


FIG. 27

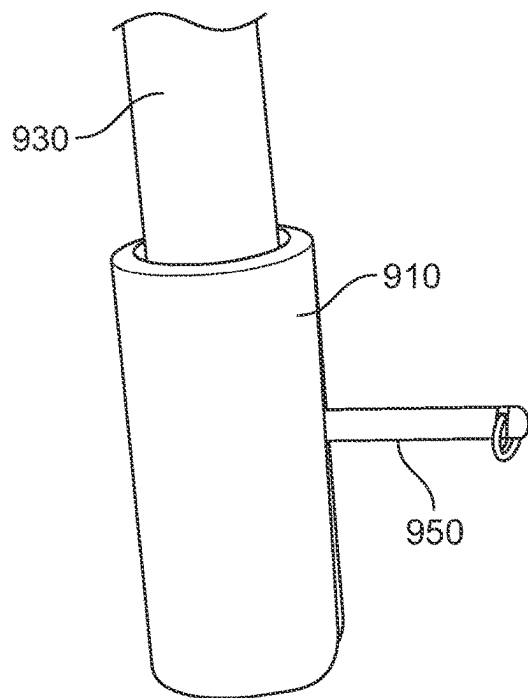


FIG. 28

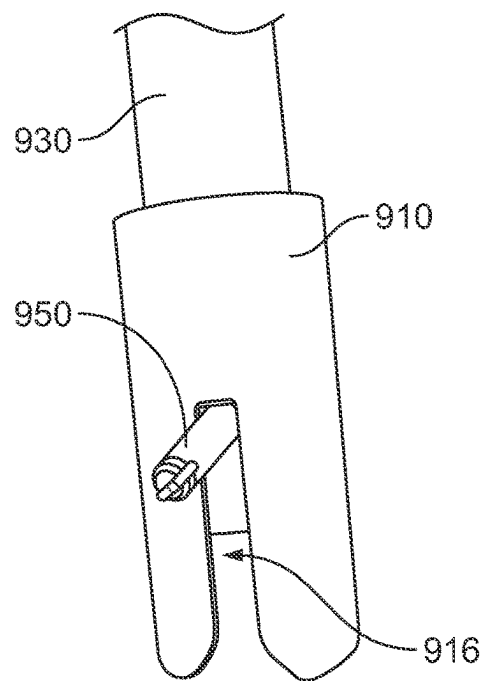


FIG. 29

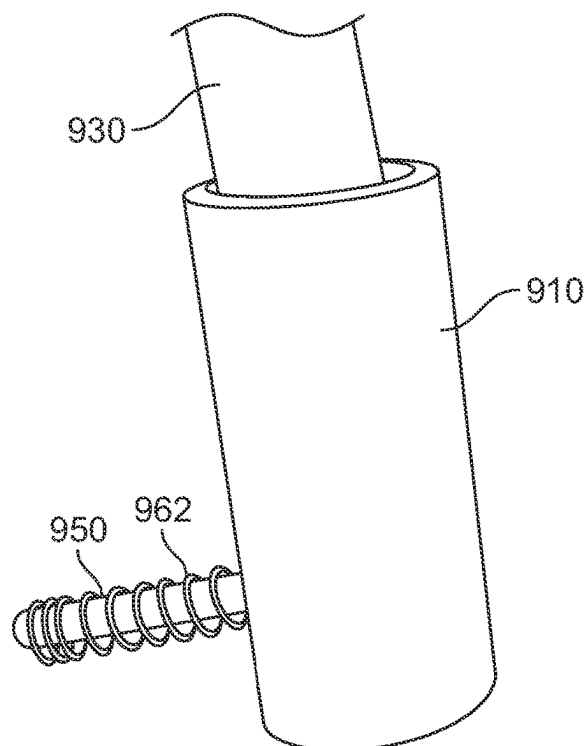


FIG. 30

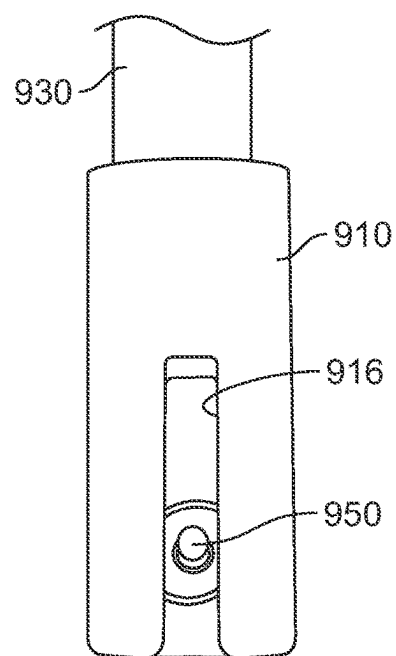


FIG. 31

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PAYLOAD DELIVERY AND DROP DEVICE FOR UNMANNED AERIAL VEHICLE FOR AUTOMATIC RELEASE UPON SURFACE CONTACT

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 17/553,118, filed Dec. 16, 2021, titled “PAYLOAD DELIVERY AND DROP DEVICE FOR UNMANNED AERIAL VEHICLE FOR AUTOMATIC RELEASE UPON SURFACE CONTACT” now U.S. Pat. No. 11,667,383, issued Jun. 6, 2023 which is a continuation of U.S. patent application Ser. No. 17/200,686, filed Mar. 12, 2021, titled “PAYLOAD DELIVERY AND DROP DEVICE FOR UNMANNED AERIAL VEHICLE FOR AUTOMATIC RELEASE UPON SURFACE CONTACT” now U.S. Pat. No. 11,225,325, issued Jan. 18, 2022, the disclosures of each of which are hereby incorporated by reference in their entireties.

BACKGROUND

The embodiments described herein relate generally to a payload delivery and drop device for an unmanned aerial vehicle (also referred to herein as a drone) or other delivery system, and more particularly, to a payload delivery and drop device for an unmanned aerial vehicle that can automatically release the payload or package in response to contact with an exterior surface (such as the ground or other delivery surface).

Known drone payload drop devices can require a user to control dropping a payload, for example via a trigger or button on a remote controller for the drone or by directing rotation of the unmanned aerial vehicle to trigger the drop. Further, known payload drop devices often attach the payload to the drop device, such as via a drop cord, drop rings, or the like. In use, portions of such attachment mechanisms and/or release mechanisms (e.g., cords, wires, rings, parachutes, or the like) are released along with the payload, leading to increased overhead costs for drone payload delivery and increased waste with respect to the released parts.

Thus, a need exists for a payload delivery and drop device that can automatically drop the payload upon contact with a delivery surface. A need also exists for a payload delivery and drop device that can release the payload in a manner that no part of the drone or release mechanism is released with the payload.

SUMMARY

Embodiments of a payload delivery and drop device for an unmanned aerial vehicle, or other delivery system, and method of its use are described herein. In some embodiments, an apparatus includes a moveable collar defining a lumen having a centerline, and a rigid bar that has a first portion and a second portion different from the first portion. The first portion of the rigid bar has a centerline and the second portion of the rigid bar has a centerline different from the centerline of the first portion. The second portion of the rigid bar is disposed within the lumen of the collar when the rigid bar is in a first position. The centerline of the first portion of the rigid bar is substantially non-parallel to the centerline of the lumen of the moveable collar when the rigid bar is in the first position. The centerline of the second

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portion of the rigid bar is substantially non-parallel to the centerline of the lumen when the rigid bar is in the second position.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a side view of an apparatus, according to an embodiment.

FIG. 2 is a front view of the apparatus of FIG. 1, in a first configuration.

FIG. 3 is a front view of the apparatus of FIG. 1, in a second configuration.

FIG. 4 is a front view and a top view of a portion of the apparatus of FIG. 1.

FIG. 5 is a front view of a portion of the apparatus of FIG. 1.

FIG. 6 is a side view of the portion of the apparatus of FIG. 5.

FIG. 7 is a side view of a portion of the apparatus of FIG. 1.

FIGS. 8-11 are front perspective views of an apparatus, according to an embodiment, in a first, second, third and fourth configuration, respectively.

FIGS. 12-13 are perspective views of rigid bars according to embodiments.

FIGS. 14-17 are schematic views of rigid bars according to embodiments.

FIG. 18 is a front perspective view of an apparatus according to an embodiment.

FIG. 19 is a schematic front view of a portion of the apparatus of FIG. 18.

FIG. 20 is a perspective view of a portion of the apparatus of FIG. 18.

FIGS. 21 and 22 are front and rear views of a portion of an apparatus according to an embodiment.

FIGS. 23 and 24 are side views of the portion of the apparatus of FIG. 21 in a first configuration and a second configuration, respectively.

FIG. 25 is an exploded view of a portion of the apparatus of FIG. 21.

FIGS. 26 and 27 are side opposing views of the portion of the apparatus of FIG. 21 with a collar portion of the apparatus in a second position, a retention pin of the apparatus in a second position, and a spring in an uncompressed state.

FIGS. 28 and 29 are side and front views, respectively, of the portion of the apparatus of FIG. 21 with the collar portion in a first position, the retention pin in a first position, and the spring (not shown) in a compressed state.

FIGS. 30 and 31 are side and front views, respectively, of the portion of the apparatus of FIG. 21 with the collar portion in a third position between its first position and its second position, the retention pin in its second position, and the spring in an uncompressed state.

DETAILED DESCRIPTION

Embodiments of a payload delivery and drop device for an unmanned aerial vehicle (also referred to herein as a drone) or other delivery system, and more particularly, to a payload delivery and drop device for an unmanned aerial vehicle that can automatically release the payload or package in response to contact with an exterior surface (such as the ground or other delivery surface) and/or without releasing any portion of the payload delivery and drop device, or a release mechanism thereof, are described herein.

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In some embodiments, an apparatus includes a moveable collar defining a lumen having a centerline, and a rigid bar that has a first portion and a second portion different from the first portion. The first portion of the rigid bar has a centerline and the second portion of the rigid bar has a centerline different from the centerline of the first portion. The second portion of the rigid bar is disposed within the lumen of the collar when the rigid bar is in a first position. The centerline of the first portion of the rigid bar is substantially non-parallel to the centerline of the lumen of the moveable collar when the rigid bar is in the first position. The centerline of the second portion of the rigid bar is substantially non-parallel to the centerline of the lumen when the rigid bar is in the second position.

In some embodiments, during use, the first portion of the rigid bar at least partially retains a payload to the apparatus when the rigid bar is in the first position, and the rigid bar permits the payload to be at least one of disposed in or released from the apparatus when the rigid bar is in the second position. The moveable collar can have a first (or retention) position in which the rigid bar is in its first position and a second (or release) position in which the rigid bar is in its second position. In some embodiments, the moveable collar prevents movement of the rigid bar from its first position to its second position when the moveable collar is in its first position. The moveable collar can move from its first position to its second position in response to a force produced when a portion of the apparatus contacts an external surface. The rigid bar can be configured to automatically move from its first position to its second position, in response to the moveable collar moving from its first position to its second position. When the moveable collar moves from its first position to its second position, the rigid bar can move from its first position to its second position in response to at least one of a weight of the payload received by the apparatus or a magnetic attraction of a portion of the rigid bar to a magnetized portion of the apparatus. The moveable collar can define a channel in fluid communication with the lumen of the collar. The first portion of the rigid bar can be extended through the channel when the rigid bar is in its first position.

In some embodiments, the apparatus also includes an elongate, rigid tube having a lumen. The rigid bar can be coupled to the elongate, rigid tube, for example, at a pivot point of the rigid bar. The pivot point of the rigid bar can be disposed within the lumen of the elongate, rigid tube.

A portion of the elongate, rigid tube can be disposed within the lumen defined by the moveable collar. The moveable collar can be slidable with respect to the elongate, rigid tube to move from its first position to its second position.

The apparatus can also include a hub configured to be coupled to a delivery system (e.g., an unmanned aerial vehicle), a plurality of radially extending arms extended from the hub, and a plurality of elongate, rigid tubes coupled to the plurality of radially extending arms such that the plurality of elongate, rigid tubes are non-parallel to the plurality of radially extending arms. The plurality of radially extending arms, the plurality of elongate, rigid tubes, and the first portion of the rigid bar (and optionally first portions of a plurality of rigid bars) can collectively define, in part, a volume within which a payload is received during use. At least one of a radially extending arm from the plurality of radially extending arms or an elongate, rigid tube from the plurality of elongate, rigid tubes can have an adjustable length.

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The moveable collar can define a channel in fluid communication with the lumen. The first portion of the rigid bar can be extended through the channel when the rigid bar is in its first position.

In some embodiments, the apparatus further includes a tube that defines a lumen, which tube can be elongate and/or substantially rigid. The moveable collar can be disposed about the tube such that the tube and the collar are substantially coaxial when the moveable collar is in a first position with respect to the tube and when the moveable collar is in a second position with respect to the tube.

In some embodiments, an apparatus according to an embodiment includes an elongate collar that has a lumen and that defines a channel in fluid communication with the lumen. The elongate collar has a first position and a second position and is configured to move from its first position to its second position in response to a force produced when a portion of the apparatus contacts an external surface during use. The apparatus also includes a rigid rod having a first portion and a second portion different than the first portion. The first portion of the rigid rod has a centerline and the second portion of the rigid rod has a centerline that intersects the centerline of the first portion of the rigid rod at a non-zero angle. The rod is configured to pivot about a pivot point disposed proximate to, for example within or at the angle of intersection, the intersection of the centerline of the second portion of the rigid rod with the centerline of the first portion of the rigid rod. The rigid rod has a first position in which the first portion of the rigid rod is extended through the channel of the elongate collar and the second portion of the rigid rod is at least partially disposed within the lumen of the elongate collar. During use, the elongate collar retains the rigid rod in its first position when the elongate collar is in its first position. Also during use, the rigid rod moves freely to its second position when the elongate collar is in its second position.

In some embodiments, the apparatus can also include a tube that defines a lumen extended at least partially there-through. The tube can be elongate and substantially rigid. A wall of the tube can define a first channel in fluid communication with the lumen of the tube and a second channel in fluid communication with the lumen of the tube. The second channel of the tube can be distinct from the first channel of the tube. The tube can be at least partially received within the lumen of the elongate collar. The elongate collar, in its first position, can be disposed about the tube at a first location of the tube such that (1) the channel of the elongate collar at least partially overlays and is in fluid communication with the first channel of the tube, and (2) the body portion of the elongate collar at least partially covers the second channel of the tube. The elongate collar, in its second position, can be disposed about the tube at a second location of the tube such that the elongate collar is spaced apart from the first channel of the tube and from the second channel of the tube.

In some embodiments, when the rigid rod is in its second position, the second portion of the rigid rod is extended through the second channel of the tube and the first portion of the rigid rod is disposed within the lumen of the tube. The tube can include a magnet configured to magnetically attract a portion of the rigid rod to retain the rigid rod in its second position, when the elongate collar is in its second position. The elongate collar can be slidably movable from its first position to its second position with respect to the tube. The elongate collar can be disposed substantially about a perimeter of the tube. The elongate collar can be coupled to a footing extended distally with respect to the collar. The

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footing, during use, can contact the external surface and can transfer a force generated by the contact to move the elongate collar from its first position to its second position.

In some embodiments, the channel of the elongate collar includes a mouth portion that is wider than an elongate portion of the channel of the elongate collar. The body or wall portion of the elongate collar that defines the mouth portion can be angled inwardly towards the elongate portion of the channel.

In some embodiments, the apparatus includes a plurality of elongate collars, and the elongate collar is a first elongate collar from the plurality of elongate collars. The plurality of elongate collars can define, in part, a perimeter of a volume of the apparatus, which volume is configured to releasably receive a payload therein. The channel of the first elongate collar can face a portion of the volume. The apparatus can also include a plurality of rigid rods, and the rigid rod can be a first rigid rod from the plurality of rigid rods. The plurality of rigid rods can define, in part, a distal boundary of a volume of the apparatus, which volume can be configured to releasably receive a payload therein.

As used in this specification and the appended claims, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, the term “a member” is intended to mean a single member or a combination of members, “a material” is intended to mean one or more materials, or a combination thereof.

As used herein, unless the context clearly dictates otherwise, the words “proximal” and “distal” refer to a direction closer to and away from, respectively, an unmanned aerial vehicle for carrying the device. Thus, for example, the end of the device configured to be attached to the unmanned aerial vehicle would be the proximal end of the device, while the end opposite the proximal end, and configured to contact a delivery surface (e.g., the ground) would be the distal end of the device. The words “proximal” and “distal” can also refer, if the context so indicates, to a direction closer to and away from, respectively, a hub of the device.

As used herein, the terms “reversible,” “reversibly,” and/or the like when used to describe a process and/or procedure generally refer to a non-destructive process or procedure that can be subsequently undone by a similar yet substantially opposed, inverse, and/or opposite non-destructive process or procedure. When used herein with respect to attachment and/or detachment of an element or assembly, a reversible attachment refers to a non-destructive, repeatable attachment and/or detachment of the element or assembly.

As used herein, the terms “about” and/or “approximately” when used in conjunction with numerical values and/or ranges generally refer to those numerical values and/or ranges near to a recited numerical value and/or range. For example, in some instances, “about 40 [units]” can mean within $\pm 25\%$ of 40 (e.g., from 30 to 50). In some instances, the terms “about” and “approximately” can mean within $\pm 10\%$ of the recited value. In other instances, the terms “about” and “approximately” can mean within $\pm 9\%$, $\pm 8\%$, $\pm 7\%$, $\pm 6\%$, $\pm 5\%$, $\pm 4\%$, $\pm 3\%$, $\pm 2\%$, $\pm 1\%$, or any other value or range of values therein or therebelow. The terms “about” and “approximately” may be used interchangeably. Furthermore, although a numerical value modified by the term “about” or “approximately” can allow for and/or otherwise encompass a tolerance of the stated numerical value, it is not intended to exclude the exact numerical value stated.

In a similar manner, term “substantially” when used in connection with, for example, a geometric relationship, a

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numerical value, and/or a range is intended to convey that the geometric relationship (or the structures described thereby), the number, and/or the range so defined is nominally the recited geometric relationship, number, and/or range. For example, two structures described herein as being “substantially non-parallel” is intended to convey that, although a non-parallel geometric relationship is desirable, some parallelism can occur in a “substantially non-parallel” arrangement. By way of another example, a structure defining a diameter that is “substantially 100 millimeters (mm)” is intended to convey that, while the recited diameter is desirable, some tolerances can occur when the volume is “substantially” the recited volume (e.g., 100 mm). Such tolerances can result from manufacturing tolerances, measurement tolerances, and/or other practical considerations (such as, for example, minute imperfections, age of a structure so defined, a pressure or a force exerted within a system, and/or the like). As described above, a suitable tolerance can be, for example, of $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, $\pm 4\%$, $\pm 5\%$, $\pm 6\%$, $\pm 7\%$, $\pm 8\%$, $\pm 9\%$, $\pm 10\%$, or more of the stated geometric construction, numerical value, and/or range. Furthermore, although a numerical value modified by the term “substantially” can allow for and/or otherwise encompass a tolerance of the stated numerical value, it is not intended to exclude the exact numerical value stated.

While numerical ranges may be provided for certain quantities, it is to be understood that these ranges can include all subranges therein. Thus, the range “from 45 to 135” includes all possible ranges therein (e.g., 46-134, 47-133, 48-132, 49-131, . . . , 89-91, etc.). Furthermore, all values within a given range may be an endpoint for the range encompassed thereby (e.g., the range 45 to 135 includes the ranges with endpoints such as 90-135, 45-90, etc.).

The apparatus described herein can be used, for example, in delivery and drop of a payload or package via an unmanned aerial vehicle, and more specifically for automatic release of the payload from the apparatus upon contact of the apparatus with an external surface, such as the ground or other delivery surface. In another example, the apparatus described herein can also or alternatively be used with, or coupled to, a manned delivery vehicle, a non-aerial delivery vehicle, a crane, robotic arm (e.g., on an aerial delivery vehicle, a land-based vehicle such as a truck, or a water-based vehicle such as a boat), or other movable delivery system.

An apparatus **100** according to an embodiment is illustrated in FIGS. 1-7. The apparatus **100** can be, for example, a payload delivery and drop device that is coupleable to an unmanned aerial vehicle (also referred to herein as a drone) and which is configured to deliver and drop a payload (or package or the like). The device **100** can be coupled to the drone (not shown) via any suitable coupling mechanism, including, for example by a cord or cable attached at one end portion to the drone and at another end portion to the payload delivery and drop device. Such a cord, cable, or other coupler can be attached (e.g. releasably attached), for example, to an eye hook **178** or the like disposed at a proximal end portion of the device **100**, such as on a hub **170** of the device **100**. In some embodiments, the hub **170** can be directly coupled to the drone (e.g., the hub can be directly bolted, screwed or otherwise attached to an underside of the drone).

The device **100** is configured to releasably receive a payload **10** within a volume **104** defined by the device. The volume **104** can be collectively defined in part, for example, by one or more of the hub **170**, one or more arms **180** that extend (e.g., radially) from the hub **170**, one or more

elongate tubes **130** coupled to one or more of the arms **180**, one or more elongate rods **190** coupled to one or more of the arms **180**, and one or more retention pins (which can be or include, for example, rigid bars or rods) **150** coupled to one or more of the elongate tubes **130**, or a combination thereof. During use, the device **100** is configured to releasably receive the payload **10** within the volume and to retain the payload **10** within the volume, as described in more detail herein. The device **100** is configured to release the payload **10** from the volume **104** in response to the device **100** contacting a delivery surface, such as the ground, a delivery pad, a rooftop, or other surface exterior to the device, as described in more detail herein.

The volume **104** can be bound, in part, on a proximal end thereof by the hub **170** and radially extending arms **180**, which can be coupled to the hub **170** in any suitable manner. For example, as shown in FIG. 1, first (or proximal) end portions of the arms **180** can be received in openings of the hub **170**. Second (or distal) end portions of at least a subset of the arms **180** are coupled to respective elongate tubes **130**, e.g., at proximal end portions **131** of the elongate tubes **130**, such that the elongate tubes are non-parallel to the radially extending arms **180**. Second (or distal) end portions of at least another subset of the arms **180** can be coupled to respective retention rods **190**, e.g., at proximal end portions thereof, such that the retention rods **190** are non-parallel to the radially extending arms **180**, as described in more detail herein.

The elongate tubes **130** each extends distally with respect to their respective arms **180** to bound, at least in part, a perimeter of the volume **104**. The elongate tubes **130** can be substantially similar or identical, and so only one tube **130** is described herein. The elongate tube **130** can be substantially rigid. As shown in FIGS. 5-6, the elongate tube **130** has a distal end portion **133** opposite the proximal end portion **131**, and defines a lumen **134** extended at least partially therethrough. The elongate tube **130** defines a centerline **CL4**, which can be perpendicular to or otherwise intersecting with a centerline of the arm **180** to which it is coupled.

The device **100** includes a collar **110** that has a lumen **114** (see, e.g., FIG. 4). The lumen **114** can be open at each end of the collar **110**. The collar **110** can be elongate, for example, in the shape of an elongate cylinder. The collar **110** operates as an actuator to release the payload from the volume **104**, as described in more detail herein. The collar **110** is coupled to the tube **130**, and more specifically, the tube **130** is at least partially received or disposed within the lumen **114** of the collar **110**. In this manner, the collar **110** is disposed about a portion of a perimeter of the elongate tube **130**. The collar **110** can have a centerline that is substantially coaxial with the centerline of the elongate tube **130**.

The collar **110** is movable from a first position (see, e.g., FIG. 3) to a second position (see, e.g., FIGS. 1-2), and vice versa. The collar **110** can be, for example, slidable with respect to the tube **130**. In some embodiments, tube **130** defines a guide channel (not shown in FIGS. 1-3, but which can be similar to a guide channel **342** shown with respect to tube **330** in FIG. 20), and a protrusion (not shown) of the collar **130** is slidably received within the guide channel. In this manner, the interaction of the protrusion of the collar and the guide channel can operate to limit movement of the collar **130** with respect to the tube in the proximal direction and the distal direction. Also in this manner, the interaction of the protrusion of the collar and the guide channel can

operate to prevent rotational movement of the collar **110** with respect to the tube **130** (e.g., when the collar **110** is in its second position).

The collar **110** is configured to move from its first position to its second position in response to a force produced when a portion of the apparatus, such as a footing **108** coupled to and extended distally with respect to the collar **110**, contacts an external surface during use. For example, when the payload delivery and drop device is in sufficiently close proximity to the delivery surface that the footing **108** contacts the delivery surface, a force produced by the contact of the footing with the delivery surface causes the footing to move in a proximal direction, and thus moves the collar **110** coupled to the footing **108** in the proximal direction to its second position. In some embodiments, release of the payload **10** from the volume **104** can be triggered by a single collar **110** being moved from its first position to its second position. In some embodiments, two or more collars **110** are moved from their respective first positions to their respective second positions to release (or drop) the payload **10** from the volume **104**. The collar **110** can include one, two (e.g., as shown in FIG. 4), three or more sleeves **111** on an outer surface of the collar **110**, which sleeve is configured to receive a portion of an elongate rod of the footing **108** therein to retain the footing to the collar **110**.

The collar **110** defines a channel **116** in fluid communication with the lumen **114**. The channel **116** of the collar can be elongate and extend substantially parallel to the centerline of the collar **110**. The channel **116** of the collar **110** faces at least a portion of the volume **104**. The channel **116** of the collar includes a mouth portion **117** that is wider than an elongate portion **119** of the channel **116**. The mouth portion **117** is defined on either side (or at least one side) of the channel **116** by a sloped, eased edge, angled or rounded corner **113** of the wall (or body) portion **112** of the collar **110** that is sloped, eased, angled or rounded inwardly towards the elongate portion **119** of the channel **116**. The wider mouth portion **117** helps to direct a first portion **152** of the rigid bar **150** into the channel **116** when the collar **110** is moved from its second position to its first position, as described in more detail herein.

The elongate collar **110**, in its first position (see, e.g., FIG. 3), is disposed about the tube **130** at a first location of the tube such that the channel **116** of the elongate collar **110** at least partially overlays and is in fluid communication with a first channel **136** defined by a wall or body portion **132** of the tube **130**. The first channel **136** of the tube is in fluid communication with the lumen **134** (see, e.g., FIG. 5-6) of the tube **130**. Thus, when the elongate collar **110** is in its first position, the channel **116** of the elongate collar **110** is also in fluid communication with the lumen **134** of the tube **130**.

The elongate collar **110**, in its first position, is also disposed about the tube **130** at the first location such that the body (or wall) portion **112** of the elongate collar **110** at least partially covers or is disposed overlaying a second channel **138** of the tube **130**. The second channel **138** is in fluid communication with the lumen **134** of the tube **130**. The second channel **138** of the tube **130** is distinct from the first channel **136** of the tube **130**. The second channel **138** can be located on the tube **130** opposite (e.g., across a diameter of tube from) the first channel **136**.

The elongate collar **110**, in its second position (see, e.g., FIGS. 1-2), is disposed about the tube **130** at a second location of the tube **130** proximal to the first location, such that the elongate collar **110** is spaced apart from (e.g., not overlaying) each of the first channel **136** and the second

channel **138** of the tube **130**. The tube **130** and the collar **110** can be substantially coaxial when the collar **110** is in its first position with respect to the tube **130** and when the moveable collar **110** is in its second position with respect to the tube **130**.

As shown in FIGS. 5-6, the rigid bar (or rod) **150** is coupled to the elongate tube **130**. For example, the rigid bar **150** can be coupled to the elongate tube **130** at a pivot point **151** (see FIG. 6), such as via a pivot pin **153** (see FIGS. 5 and 7) extended at least partially through the wall portion **132** of the tube **130**. The pivot point **151** of the rigid bar **150**, and optionally a portion of pivot pin **153**, can be disposed within the lumen **134** of the tube **130**. The rigid bar **150** has a first portion **152** and a second portion **156** different from the first portion. The first portion **152** of the rigid bar **150** has a centerline CL2 and the second portion **156** of the rigid bar has a centerline CL3 different from the centerline CL2 of the first portion. In some embodiments, for example, the first and second portions **152**, **156** of the rigid bar **150** can form a substantially L or V-shape. The centerline CL3 of the second portion **156** can intersect the centerline CL2 of the first portion **152** of the rigid bar **150** at a non-zero angle. For example, at an angle within the range of about 45 degrees to about 135 degrees, or within the range of about 80 degrees to about 100 degrees, or at about 90 degrees. The pivot point **151** about which the rigid bar (or rod) **150** is configured to pivot can be disposed within the angle of intersection of the centerlines CL2, CL3 of the first and second portions **152**, **156** of the rigid bar **150**, or it can be disposed at the intersection of the centerlines CL2, CL3.

The rigid bar **150** has a first (or retention) position (see, e.g., FIG. 3) and a second (or release) position (see, e.g., FIGS. 1-2 and 5). The second portion **156** of the rigid bar **150** is at least partially disposed within the lumen **114** of the collar **110**, and also within the lumen **134** of the tube **130**, when the rigid bar **150** is in its first position. Also when the rigid bar **150** is in its first position, the centerline CL2 of the first portion **152** of the rigid bar **150** is substantially non-parallel to the centerline CL1 of the lumen **114** of the moveable collar **110** (and to the centerline CL4 of the lumen **134** of the tube **130**). For example, the first portion **152** of the rigid bar **150** can be substantially perpendicular to the centerline CL4 of the tube **130** and/or the collar **110**, when the rigid bar **150** is in its first position. The first portion **152** of the rigid bar **150** is extended through the first channel **136** of the tube **130** and through the channel **116** of the collar **110**, when the rigid bar **150** is in its first position. In this manner, the first portion **152** of the rigid bar **150** defines a lower boundary of the volume **104**, when the rigid bar **150** is in its first position. As such, during use, the first portion **152** of the rigid bar **150** can at least partially retain the payload **10** to the apparatus **100**, e.g., within the volume **104**, when the rigid bar **150** is in the first position.

The rigid bar **150** moves from its first position to its second position by pivoting about the pivot point **151**. When the rigid bar **150** is in its second position, the centerline CL3 of the second portion **156** of the rigid bar **150** is substantially non-parallel to the centerline CL1 of the lumen **114**. For example, the second portion **156** of the rigid bar **150** can be substantially perpendicular to the centerline CL4 of the tube **130** and/or the collar **110**, when the rigid bar **150** is in its second position. The rigid bar **150** permits the payload **10** to be at least one of disposed (or received) in, or released from, the device **100** when the rigid bar **150** is in the second position.

The collar **110** has a first position in which the rigid bar **150** is in its first position and a second position in which the

rigid bar **150** is in its second position. In its first position, the moveable collar **110** prevents movement of the rigid bar **150** from its first position to its second position. Said another way, during use, the collar **110** retains the rigid bar **150** in its first position when the collar **110** is in its first position. The moveable collar **110** does not, however, prevent movement of the rigid bar **150** from its first position to its second position when the collar **110** is in its second position. In other words, when the collar **110** is moved proximally away from its first position and to or towards its second position, the collar **110** is no longer disposed over the second channel **138** of the tube **130**, and so the rigid bar (or rod) **150** can pivot about its pivot axis **151** to move the rigid bar **150** to its second position. As such, during use, the rigid bar **150** can move freely to its second position when the collar **110** is in its second position. When the rigid bar **150** is in its second position, the second portion **156** of the rigid bar **150** is extended through the second channel **138** of the tube **130**. Also when the rigid bar **150** is in its second position, the first portion **152** of the rigid bar **150** is disposed within the lumen **134** of the tube **130**. The rigid bar **150** can be configured to automatically move from its first position to its second position, in response to the moveable collar **110** moving from its first position to its second position.

When the collar **110** moves from its first position to its second position, the rigid bar **150** can move from its first position to its second position in response to at least one of a weight of the payload **10** received by the apparatus **100** (e.g., in the volume **104**) and/or a magnetic attraction of a portion of the rigid bar **150** to a magnetized portion of the device **100**. In some embodiments, for example, the rigid bar **150** is constructed, at least in part, of a ferromagnetic metal and the tube **130** includes a magnet **142** (schematically illustrated in FIG. 5) positioned with respect to the tube such that the magnetic attraction of the rigid bar **150** causes the rigid bar **150** to tend towards its second position. The magnet **142** can be located, for example, within the lumen **134**, within the first channel **136** or the second channel **138** (e.g., towards a distal end portion thereof), on a surface of the tube **130** proximate the distal end portion of one of the channels **136**, **138**, or at a distal end portion of the tube **130**. In some embodiments, the rigid bar **150** can be constructed of magnesium or other suitable material, and can include a steel tip portion that is configured to magnetically engage with the magnet **142**. In some embodiments, the first portion **152** of the rigid bar **150** has a length greater than a length of the second portion **156** of the rigid bar **150**. In this manner, the first portion **152** of the rigid bar **150** can extend more closely to the magnet **142** than the second portion **156** of the rigid bar.

The payload delivery and drop device **100** as shown in FIGS. 1-3 includes two tubes **130**, each with a moveable, elongate collar **110** and rigid bar (or rod) **150** coupled thereto, which tubes **130** (and thus the collars **110** and rigid bars **150**) are positioned at opposing sides of the hub **170**. The device **100** also optionally includes elongate rods **190** coupled to a subset (e.g., two as shown in FIGS. 1-3) of arms **182** such that the elongate rods **190** extend distally with respect to the arms **182**. The elongate rods **190** can define, in part, a portion of the perimeter of the volume **104**. The elongate rods **190** prevent the payload **10** received in the volume **104** from moving laterally out of the volume between the elongate tubes **130**. In some embodiments, one or more of the rods **190** has an adjustable length.

While various embodiments have been described above, it should be understood that they have been presented by way of example only, and not limitation. Where schematics

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and/or embodiments described above indicate certain components arranged in certain orientations or positions, the arrangement of components may be modified. While the embodiments have been particularly shown and described, it will be understood that various changes in form and details may be made. Although various embodiments have been described as having particular features and/or combinations of components, other embodiments are possible having any combination or sub-combination of any features and/or components from any of the embodiments described herein. Further, although methods have been described herein in reference to a specific embodiment, the methods can be executed using any suitable device embodiment described herein.

For example, while the arms **180** and tubes **130** are shown in FIGS. 1-3 as being substantially perpendicular (e.g., with the arms extended along a substantially horizontal axis and with the tubes **130** being extended along a substantially vertical axis, relative to a ground surface when the drone is positioned or flying parallel to the ground), in other embodiments, the arms and tubes can define a different interior (i.e., facing the volume) angle. For example, in other embodiments, an arm and a tube can be coupled to form an interior angle within the range of about 45 degrees to about 135 degrees. In such embodiments, the arm can have a centerline that intersects the horizontal axis, the tube can have a centerline that intersects the vertical axis, or both.

In another example, although the payload delivery and drop device **100** is shown and described herein as including two tubes **130** each with a collar **110** and rigid bar **150** configured to selectively retain the payload to the device, in other embodiments, the delivery device **100** can include any suitable number of tubes, collars and/or rigid bars, including, but not limited to, one, three, four or more tubes, collars and/or rigid bars. For example, a payload delivery and drop device **200** according to an embodiment is shown in FIGS. 8-10. The device **200** can be similar in many respects, or identical to, the payload delivery and drop devices described herein (e.g., device **100**, **300**) and can include features similar in many respects, or identical to features of the payload delivery and drop devices described herein. As such, the device **200** and its features are not described in detail herein.

The device **200** includes a hub **270**, a set of arms **280** extending radially from the hub **270**, and a set of four elongate tubes **230** coupled to and extending distally with respect to the arms **280**. Each tube **230** is received within a lumen **214** defined by a collar **230**. Each tube **230** also has a rigid bar (or rod) **250** pivotally coupled thereto, such that the rigid bar **250** can move from a first position in which the rigid bar **250** retains a payload **10** within a volume **204** of the apparatus to a second position in which the payload **10** can be released from the volume (or optionally, loaded into the volume).

In use, the rigid rods **250** are moved to their second position, such that a first portion **252** of each rod **250** is disposed within a lumen (not shown) of each tube **230**. The rigid rods **250** can be retained in the second position by a magnet (not shown) coupled to the tubes **230**, or a magnetized portion of the tubes **230**. A payload **10** is loaded into the volume **204** of the device **200**. At least one moveable collar **210** is moved from its second position to its first position, such that at least one rigid rod **250** is moved from its second position to its first position. In this manner, the payload **10** is retained in the volume **204** of the apparatus, collectively, by the rigid rods **250**, elongate tubes **230**, and arms **280** (and optionally, the hub **270**). An unmanned aerial

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vehicle (drone; not shown) can be coupled to the device **200**, such as by attaching the vehicle to the eye hook or loop **278** of the device **200**. The drone can be flown to the desired delivery location and optionally vertically aligned with a desired delivery or exterior surface (e.g., the ground, concrete, a rooftop, a delivery pad, or the like). The drone lowers the device **200** towards the delivery surface such that a footing **208** coupled to the collar **210** (or a distal end portion of the collar in some embodiments) engages the delivery surface. A force produced in response to contact of the footing **208** (or collar **210**) with the delivery surface causes the collar **210** to move in a proximal direction from its first position to its second position, thereby releasing its restraint on the pivotable rigid rod **250**. The rigid rod **250** moves from its second position to its first position, which releases the lower boundary of the volume **204** within which the payload **10** is received and so the payload **10** is dropped or released from the volume **204**. In this manner, the device **200** is configured to deliver the payload **10** while the delivery vehicle or system (e.g., a drone, or other delivery vehicle or system described herein) remains in the air during release of the payload **10**. The payload **10** can be released without any release or attachment mechanism, or other portion of the device **200** (e.g., a tether, chain, strap, or the like) remaining attached to the payload **10** or otherwise being separated from the device **200** after release of the payload **10**. In other words, in some embodiments, only the payload **10**, and no other component, is released from the device **200** during delivery of the payload **10**. The device **200** can then be optionally returned, e.g., via the drone, to for example a loading or distribution center, and can be loaded with another payload so that another the delivery and drop process can be executed or repeated.

FIGS. 8-11 depict a payload release sequence, with the payload **10** retained in the device **200** shown in FIG. 8, with footings **208** of the device **200** contacting an exterior surface **20** thereby initiating movement of the collars **210** towards their second positions and the rigid rods **250** towards their second positions shown in FIG. 9, with the collars **210** and rigid rods **250** in an intermediate position between their first positions and their second positions shown in FIG. 10, and with the collars **210** and rigid rods **250** in their respective second positions and the payload released from the volume **204** shown in FIG. 11.

In yet another example, while the tubes **130** and rods **190**, or the tubes **230**, are shown as being substantially equidistantly spaced apart (e.g., such that the volume has a substantially square perimeter shape, from a top view), in other embodiments, a payload delivery and drop device can include tubes and/or retention rods having a different configuration. For example, the device can include tubes and/or retention rods spaced such that the volume has a substantially rectangular perimeter shape (from a top view). In another example, the device can have less than four or more than four such tubes and/or retention rods, which can be spaced such that the volume has a triangular, ellipsoidal, circular, or polygonal perimeter shape (from a top view).

In still another example, although the rigid bar **150** has been shown and described herein as having first and second portions with substantially linear centerlines that intersect at a pivot point **151**, a pivotable rigid bar or rod can have a different configuration. For example, pivotable rigid bars according to embodiments are shown in FIGS. 12-15. FIG. 12 shows a set of four rigid rods **350**. Each rigid rod **350** includes a first end portion **352** and a second end portion **356** each of which is substantially linear, at least proximate to free ends thereof. The first end portion **352** and the second

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end portion **356** define an interior angle, and the rigid rod **350** includes a pivot plate **363** disposed within the interior angle. The pivot plate **363** can have a substantially concave curvature. The rigid rod **350** is configured to pivot about a pivot point **351** defined by the pivot plate **364**. The pivot point **351** is positioned inside an angle **ANG** of intersection between a centerline **CL5** of the first portion **352** of the rigid rod **350** and a centerline **CL6** of the second portion **356** of the rigid rod **350**. A pivot pin **353** can be disposed through an opening (not shown) at the pivot point **351**, such as to couple the rigid rod **350** to an elongate tube as described herein.

In FIG. **13**, a pivotable rigid rod **450** includes a first portion **452** and a second portion **456**. In a manner similar to that described with respect to rigid rod **350**, the rod **450** can include a pivot plate **463** disposed within an interior angle formed between the first and second portions **452**, **456** of the rod **450** and can have a pivot point **451** within the plate **463**. A distal end portion **454** of the first portion **452** of the rod **450** includes a crossbar **462** coupled thereto. The crossbar **462** is configured to provide additional support and stability to a payload received in a volume of a payload delivery and drop device, in addition to that provided by the first portion **452** of the rigid rod **450**. Although the crossbar **462** is shown in FIG. **13** as being substantially perpendicular to the first portion **452** of the rigid rod **450**, in some embodiments, the cross bar can be differently configured, such as in the shape of a C, E, U, V, W, Y, or other suitable shape, including a non-planar shape. Although the retention pin has been shown and described as rigid bars or rods **150**, **250**, **350**, **450** that include substantially linear first and second portions, in some embodiments, one or more of the first or second portions of a rigid rod can be differently configured or the device can include a retention pin differently configured. For example, as shown in FIG. **14**, a rigid rod **550** according to an embodiment includes a substantially linear first portion **552** (with an optionally non-linear foot at a free end thereof) and a second portion **556** with a varying thickness along a length thereof, and a pivot point **551** therebetween. In another example, as shown in FIG. **15**, a retention pin **650** according to an embodiment can include a rigid first portion **652** that is substantially linear (with an optionally non-linear foot at a free end thereof) and a second portion **656** that is curved (e.g., in a substantially convex curve) and a third portion **658** that connects an end **657** of the second portion **656** to a pivot point **651**. The second portion **656** can be configured to elastically deform to permit the collar to slide over the second portion **656** during use, but still have sufficient rigidity to maintain a resistance against an inner surface of the collar when the second portion is elastically deformed and disposed within the collar.

In yet another example, as shown in FIG. **16**, a rod **750** can include a pivot point **751**, a first portion **752** (with an optionally non-linear foot at a free end thereof) and a second portion **756** that defines a shoulder **757** at its free end. A portion of the shoulder **757** can extend from the second channel (e.g., second channel **138**) of the tube (e.g., tube **130**) when the rod **750** is in its first position. In still another example, as shown in FIG. **17**, a retention pin (or rod) **850** includes pivot point **851**, a first portion **852**, a second portion **856** that has a greater thickness (or cross-sectional diameter) than the first portion **852**, and optionally a third portion **858** that connects the second portion **856** to a free end of the first portion **852**. The first portion **852** can optionally have a non-linear foot or hook portion at its free end.

Although the rigid rods (e.g., rigid bars or rods **150**, **250**, **350**, **450**, **550**, **650**) have been shown and described herein

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as including a first portion and a second portion, each having a length, in other embodiments, one or both of the first portion and the second portion of the rigid rod can have a different (longer or shorter) length than the first and second portions described herein, and the first portion can have a different (longer or shorter) length than the second portion.

Although the footing **108**, **208** has been shown and described herein as being a substantially continuous rod coupled at opposing ends thereof to the collar **110**, **210**, in other embodiments, a payload delivery and drop device can include a footing that is differently configured. For example, referring to FIGS. **18-20**, a payload delivery and drop device **300** according to an embodiment is shown. The device **300** can include features similar in many respects, or identical to, features of other devices described herein (e.g., device **100**, **200**), and so the features of device **300** are not described in detail herein. The payload delivery and drop device **300** includes a hub **370**, a set of radially extending arms **380**, a set of elongate tubes **330** extended distally from the arms **380**, a collar **310** defining a lumen **314** within which a portion of the tube **330** is received and a channel **316**, and a rigid bar **350** pivotally coupled to the tube **330** for selectively retaining a payload to the device **300**, as described with respect to other devices herein. The device **300** includes a footing **308** coupled to the collar **310**. The footing **308** is shown as including two elongate rods, a proximal end of each of which is coupled to an outer surface of the collar **310** via a flexible clamp (or the like) (see, e.g., FIG. **20**) and a distal end of each of which is bent to form a foot **309** configured to contact the exterior or delivery surface. Although each collar **310** is shown and described as including a footing **308** having two elongate rods with feet **309**, in other embodiments, the footing can include one rod with a foot, or three or more rods.

Although the device **100** has been shown and described herein as including a magnet **142** configured to influence the rigid rod **150** towards its second position, in other embodiments, the device can be differently configured. For example, in some embodiments, a device can include a spring configured to bias or return the rigid rod to its first position absent a retaining force.

Although the devices herein (e.g., device **100**, **200**, **300**) have been shown and described herein as including a retention pin (e.g., a rigid bar or rod) with a first portion that is non-parallel to a second portion thereof, in other embodiments, a device according to an embodiment can have a retention pin that is differently configured. For example, referring to FIGS. **21-31**, a device **900** according to an embodiment is configured to releasably receive a payload **10** within a volume (not shown) defined by the device. The volume can be collectively defined in part, for example, by one or more elongate tubes **930**, one or more collars **910** moveably disposed about the one or more elongate tubes **930**, and one or more retention pins (which can be or include, for example, rigid bars or rods) **950**. The device **900** can be similar or identical in many respects, and include one or more features similar or identical in many respects, to other devices described herein (e.g., device **100**, **200**, **300**). As such, many features of the device **900** are not shown or described in detail herein.

The retention pin **950** includes a first end portion **952** and a second end portion **956**, and can be substantially linear between its second end portion **956** and its first end portion **952**. The retention pin **950** can be at least partially disposed within a lumen (not shown) of the tube **930**, which lumen can be extended along at least a portion of the centerline of the tube, such that a centerline **CL8** of the retention pin **950**

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is non-parallel to (and, optionally, transverse or orthogonal to) a centerline CL7 of the tube 930. In some embodiments, the tube 930 also or alternatively defines a lumen 942 that is non-parallel to (and, optionally, transverse or orthogonal to) the centerline CL7 of the tube 930. This lumen 942 can be

coaxial with the retention pin 950. The retention pin 950 can include a spring 962 disposed about at least a portion of the retention pin 950 between its first and second end portions 952, 954. The spring 962 can be coupled to the retention pin 950, for example to at least one of the first or second end portion 952, 954 of the retention pin 950. The spring 962 can be disposed about the retention pin 950 between a rubber wheel 968 coupled to the second end portion 956 of the retention pin and a washer 964 coupled to the first end portion 952 of the retention pin 950 to keep the spring 962 disposed on the retention pin 950. In this manner, the spring 962 can move freely about the retention pin 950 without the spring being removed from being disposed about the retention pin 950. The rubber wheel 968 can be coupled to the second end portion 956 of the retention pin 950, for example, such that the rubber wheel 968 overlays an end face of the second end portion 956 of the retention pin 950 or such that the rubber wheel 968 is disposed about the second end portion 956 of the retention pin 950. In this manner, the second end portion 956 is prevented from being fully withdrawn into or through the tube 930 during use, as described in more detail herein. The washer 964 and a retaining pin 966 configured to couple the washer 964 to the retention pin 950 can be coupled to the first end portion 952 of the retention pin 950. The washer 964 and pin 966 collectively operate to prevent the first end portion 952 of the retention pin 950 from being withdrawn fully into or through the tube 930 during use, as described in more detail herein.

The retention pin 950 is moveable with respect to the tube 930. For example, the retention pin 950 has a position (e.g., a first position) (see, e.g., FIG. 24) in which the second end portion 956 of the pin is substantially disposed within the tube 930 (and/or lumen 942) and the first end portion 952 of the retention pin 950 is extended from (disposed outside of) the tube 930 (and/or lumen 942) on a first side of the tube 930, and another (e.g., second) position (see, e.g., FIG. 23) in which the second end portion 956 of the retention pin 950 is extended from (disposed outside of) a second side of the tube 930 (and/or lumen 942) and the first end portion 952 of the retention pin 950 is substantially disposed within the tube 930 (and/or lumen 942).

When the retention pin 950 is in its first position, the spring 962 is in a compressed state. For example, the spring 962 can be compressed between the rubber wheel 968 and the tube 930. When the retention pin 950 is in its second position, the spring 962 disposed about the retention pin 950 is in a substantially uncompressed state. When the spring 962 is in the substantially uncompressed state, for example, the spring can be uncompressed or compressed slightly, such as to an extent that the retention pin 950 does not move from its second position towards its first position absent application of an external force. The spring 962 can be biased towards the uncompressed state. In this manner, when the retention pin 950 is in its first position, the spring 962 causes the retention pin 950 to move to its first position in the absence of (or upon removal of) a force retaining the retention pin 950 in its first position. As such, the retention pin 950 is biased towards its second position.

When the retention pin 950 is in its first position, the first end portion 952 of the retention pin 950 defines a lower boundary of the volume of the device 900. As such, during

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use, the first end portion 952 of the retention pin 950 can at least partially retain the payload 10 to the apparatus 900, e.g., within the volume (which, for example, is to the right of the tube 930 and collar 910 and also above the retention pin 950 in the views shown in FIGS. 24 and 28), when the retention pin 950 is in the first position.

The retention pin 950 is moved to its first position by pushing the second end portion 956 towards the second side of the tube 930 and/or pulling the first end portion 952 of the retention pin 950 away from the first side of the tube 930. A collar 910 that is disposed about and slideably moveable with respect to the tube 930 can be moved into a first position in which a wall portion of a second side portion of the collar 910 overlays the second end portion 956 of the retention pin 950 and the rubber wheel 968. In this manner, the collar 910 in its first position prevents the spring from decompressing and moving the retention pin 950 to its second position, thereby retaining the retention pin 950 in its first position. A first side portion of the collar 910 defines a channel 916. The first end portion 952 of the retention pin 950 extends through the channel 916 of the collar 910 (see, e.g., FIGS. 28-29) when the retention pin 950 is retained in its first position by the collar 910.

In use, during delivery of a payload by the device 900, the collar 910 is moved in a proximal direction from its first position towards a second position in which the collar 910 is proximally spaced apart from the retention pin 950 and/or lumen 942 (see, e.g., FIGS. 26-27). In some embodiments, the collar 910 can be moved from its first position to a third position between its first and second positions (see, e.g., FIGS. 30-31) with respect to the tube 930, during release of the payload. The collar 910 can be moved from its first position to or towards its second position or its third position in any suitable manner described herein, such as with respect to collars 110, 210 310. In each of the second and third positions, the wall portion of the collar 910 no longer overlays the rubber wheel 968 and/or second end portion 956 of the retention pin 950, thereby permitting the spring 962 to move to its uncompressed state and the retention pin 950 to also move to its second position (e.g., and thereby release the payload).

The collar 910 can defined a notched or cut-out portion 911 on the second side portion of the collar 910, as shown in FIG. 27, though which the second end portion 956 of the retention pin 950 and the spring 962 can be extended when the retention pin 950 is moved to its second position and the collar 910 is in its third position. In its third position, a wall portion of the collar 910 can be disposed about a portion of the tube 930 defining the lumen 942, but not simultaneously overlay the rubber wheel 968 and/or second end portion 956 of the retention pin 950. Thus, the notched portion 911 enables the retention pin 950 to move from its first position to its second position, without the entire length of the collar 910 having to be moved proximally away from the retention pin 950. In other words, because of the notched portion 911 of the collar 910, the collar 910 need not be moved fully from its first position to its second position to permit the spring 962 to cause the retention pin 950 to move to its second position, thereby releasing the payload.

Although device 900 is described as including a spring 962, in some embodiments, a device according to an embodiment can include both a magnet and a spring, as described herein, or neither.

Although devices herein (e.g., device 100, 200, 300) have been shown and described herein as including a hub (e.g., hub 170, 270, 370) and radially extending arms (e.g., arms 180, 182, 280, 380), in other embodiments, the device can

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be differently configured. For example, in some embodiments, a payload delivery and drop device can include a hub in the form of a plate, ring, square, rectangle, or other polygon (each having a solid profile in some embodiments, and each having an opening inside the outer perimeter in other embodiments). In some embodiments, the hub has a perimeter shaped and sized to permit the tubes (e.g., tubes 130, 230, 330) and/or retention rods (e.g., retention rods 190) to be directly coupled to the hub. In this manner, for example, a device can be devoid of arms that extend radially from the hub.

The devices described herein, and various components thereof, can be constructed of any suitable material. For example, in some embodiments, the collar, tube, arm, hub, or other portion of a device herein can be constructed of nylon or another suitable material.

The specific configurations of the various components described herein can also be varied. For example, the size and specific shape of the various components can be different from the embodiments shown, while still providing the functions as described herein. Additionally, the relative size of various components of the devices shown and described herein with respect to the size of other components of the devices are not necessarily to scale.

For example, although the tubes and collars are each shown and described herein as being substantially cylindrical, in other embodiments, however, a tube and/or collar can have any suitable perimeter shape that permits the collar to translate, or move with respect to the tube, as described herein for automatic release of the rigid bar from its first position to its second position. For example, the tube and/or collar can have an elliptical, square, rectangular, or polygonal perimeter shape, which shape can, for example, limit rotational movement between the collar and the tube.

Similarly, where methods and/or events described above indicate certain events and/or procedures occurring in certain order, the ordering of certain events and/or procedures may be modified. While the embodiments have been particularly shown and described, it will be understood that various changes in form and details may be made.

What is claimed is:

1. An apparatus, comprising:

an aerial vehicle; and

a payload release device coupled to an underside of the aerial vehicle, the payload release device having a first configuration and a second configuration,

while the aerial vehicle is in flight and the payload release device is in the first configuration, the payload release device configured to at least partially retain a payload, the payload release device configured to move from the first configuration to the second configuration in response to the payload release device being lowered into contact with an external surface,

while the payload release device is in the second configuration, the payload release device configured to not retain the payload,

the payload release device includes a plurality of elongate members that collectively define, at least in part, a volume within which a payload is retained when the payload release device is in the first configuration,

each elongate member from the plurality of elongate members including a collar and a rigid bar, the rigid bar of each elongate member from the plurality of elongate members configured to limit downward movement of the payload within the volume when the payload release device is in the first configuration,

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the rigid bar of each elongate member from the plurality of elongate members configured to not limit downward movement of the payload within the volume when the payload release device is in the second configuration.

2. The apparatus of claim 1, wherein:

the plurality of elongate members configured to limit a range of motion of the payload while retained within the volume.

3. The apparatus of claim 1, wherein the payload release device is configured to retain the payload within the volume without a tether when the payload release device is in the first configuration.

4. An apparatus, comprising:

a payload release device configured to be coupled to an underside of an aerial vehicle, the payload release device having a plurality of members collectively defining at least in part a volume within which a payload is at least partially retained when the payload release device is in a first position,

the payload release device having a range of motion between the first position in which the payload is at least partially retained and a second position in which the payload is not retained,

the payload release device configured to move from the first position to the second position in response to at least a portion of the payload release device contacting an external surface during lowering of the payload release device,

each elongate member from the plurality of members including a collar and a rigid bar, the rigid bar of each elongate member from the plurality of members configured to limit downward movement of the payload within the volume when the payload release device is in the first position,

the rigid bar of each elongate member from the plurality of members configured to not limit downward movement of the payload within the volume when the payload release device is in the second position.

5. The apparatus of claim 4, wherein:

the plurality of members is configured to moveably retain the payload within the volume when the payload release device is in the first position,

the plurality of members configured to limit a range of motion of the payload while retained within the volume.

6. The apparatus of claim 4, wherein the payload release device is configured to retain the payload within the volume without a tether when the payload release device is in the first position.

7. A method, comprising:

retaining a payload at a payload release device while an aerial vehicle is in flight and the payload release device is in a first configuration, the payload release device coupled to an underside of the aerial vehicle, the payload release device including a plurality of elongate members that collectively define, at least in part, a volume within which a payload is retained when the payload release device is in the first configuration, each elongate member from the plurality of elongate members including a collar and a rigid bar, the rigid bar of each elongate member from the plurality of elongate members configured to limit downward movement of the payload within the volume when the payload release device is in the first configuration;

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moving the payload release device from the first configuration to a second configuration in response to the payload release device being lowered into contact with an external surface; and

not retaining the payload to the payload release device 5 while the payload release device is in the second configuration, the rigid bar of each elongate member from the plurality of elongate members configured to not limit downward movement of the payload within the volume when the payload release device is in the second configuration. 10

8. The method of claim 7, wherein:

the plurality of elongate members configured to limit a range of motion of the payload while retained within the volume. 15

9. The method of claim 7, wherein the payload release device is configured to retain the payload within the volume without a tether when the payload release device is in the first configuration.

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10. The method of claim 7, wherein the payload release device includes:

a hub configured to be coupled to the aerial vehicle;
a plurality of radially extending arms extended from the hub; and

a plurality of elongate, rigid tubes coupled to the plurality of radially extending arms such that the plurality of elongate, rigid tubes is non-parallel to the plurality of radially extending arms;

the plurality of radially extending arms, the plurality of elongate, rigid tubes, and the plurality of elongate members collectively define, in part, the volume within which the payload is received when the payload release device is in the first configuration.

11. The method of claim 10, wherein at least one of a radially extending arm from the plurality of radially extending arms or an elongate, rigid tube from the plurality of elongate, rigid tubes has an adjustable length.

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